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Denver Regional Transportation District (DENVER RTD)

RTD Denver EMU Door System

Project # 7230

Door Control System

Software Quality Assurance Plan (SQAP)

Contract Number	-	CDRL Number	-
		Customer Ref. Number	-

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0	2011-05-02	Slavica Lezaic		Original issue.

Preface

This Software Quality Assurance Plan (SQAP) complies with the requirements of the IEEE 730-2002, IEEE Standard for Software Quality Assurance Plans, approved on 2002-09-23, EN50128-2001 and IEEE 1558-2004, Standard for Software Documentation

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1 PURPOSE AND SCOPE

1.1 Purpose

This Software Quality Assurance Plan (SQAP) addresses the quality planning for developing and integrating the Door System software for RTD Denver project

Updates to this SQAP will address the RTD Denver Door System Software changes and upgrades.

This SQAP identifies applicable policies, requirements, and standards for Door System software development. It defines the organization, resources, and processes to be followed for software quality assurance activities necessary to accomplish the development.

The plan is intended for Vapor Stone Rail Systems Inc. (hereafter referred to as VSRS) software project management, VSRS development team, VSRS maintenance team, Denver Regional Transportation District (hereafter referred to as DENVER RTD) and Hyundai Rotem.

1.2 Scope

This SQAP applies to the software development to be performed for the Door Control System as part of the RTD Denver project. It is applicable particularly to the software items listed in Table 1-1 during the complete life cycle of the RTD Denver project.

The SQA's objective is to ensure that the activities described in the planning documents (i.e. SPMP, SQAP, SCMP and SVVPs) are performed and that the guidelines defined in the standards and the company policies and processes are followed throughout the entire life cycle of the project. Additionally, the SQA identifies the non-conformances related to the work products and the software development process.

1.3 Work Packages and Dependencies

Work packages and tasks that must be completed in order to satisfy the project agreement are presented in the VSRS SEP document #CMM-01. All activities are classified in ascending project progress. In general, the input of an activity has a dependency with the result/output from the previous activity. Detailed dependency relations and activities that can be performed in parallel are clearly identified in the VSRS SEP document.

Figure 1-1 depicts the hardware and software work package dependencies.

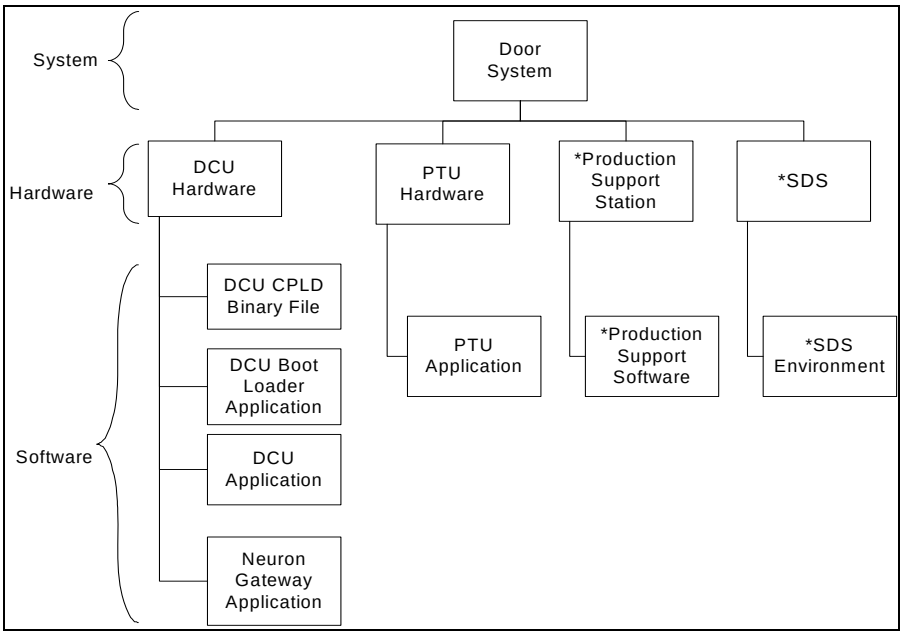

Figure 1-1: Hardware and Software Dependencies

*Items are not SCIs/deliverable for this project and will be used internally. Their purpose is described in the sections 1.3.1 and 1.3.2:

Table 1-1: Software Categorization

SCI	Category	SIL
DCU Boot Loader Application	A	0
DCU Application	A	2
DCU CPLD Binary File	A	2
Neuron Gateway Application	A	0
PTU Application	A	0

The categories are defined as follow:

- A - Application-specific software, developed or adapted specifically for this Contract;
- B - Application-independent software, that is, fixed system software that is used in multiple applications (e.g., operating systems), or software that is encapsulated in a replaceable component (e.g., intelligent power modules).

1.3.1 Production Support Station (PSS)

The PSS hosts the Production Support software which controls the following functions:

- Upload the DCU Boot Loader Application and DCU Application into the DCU for the Production purposes;
- Upload the DCU Neuron Gateway Application into the Neuron Gateway processor;
- Test the hardware using internal test tools, namely the CPU Test and RTP Tool;
- Allows to program the CPLD;
- Allows monitoring of DCU while in the Burn-In environment chambers.

1.3.2 Software Development Station (SDS)

Software Development Station (SDS) hosts the Development Software. It allows to:

- Maintain (edit, modify and recompile) all above defined software;
- Maintain (edit and print) the documentation of all above software, including attachments and embedded objects, but excluding AutoCAD and Solid-Edge drawings;
- Maintain (edit, modify and recompile) programmable devices;
- Create the media (e.g. floppy disks, CD-ROM) and labels required for release of the maintained items, if any;
- Upload the DCU Boot Loader Application and DCU Application into the DCU for the Development purposes;
- Upload the DCU Neuron Gateway Application into the Neuron Gateway processor;
- Allows to program the CPLD.

1.4 Software Configuration Items Covered

The following provides an overview of the system.

1.4.1 DCU Door Control Unit

The DCU Hardware hosts the DCU Application, the Neuron Gateway Application, the DCU Boot Loader Application and DCU CPLD Binary File.

1.4.1.1 DCU Application controls the following functions:

The DCU Application is part of the DCU to control the following functions:

- Door control;
- Interface with input and output control;
- Monitoring Operations;
- Data Logging Operations;
- Diagnostics;
- Troubleshooting;
- PTU Communication (host side);
- Neuron Gateway communication.

1.4.1.2 DCU Boot Loader controls the following functions:

- Initial CPU and peripheral initialization,
- Initial self test and memory integrity checks,
- DCU Application Start-up,
- DCU Application updates through PTU interface port.

1.4.1.3 **The Neuron Gateway CPU** is used as a gateway between Car Network and Monitoring and Control CPU.

1.4.1.4 DCU CPLD Binary File is part of the DCU to control the following functions

- Hardware based signal conditioning
- Emergency mode
- Input reading
- H-bridge control
- CPLD updates through JTAG Interface

1.4.2 PTU

The PTU host the PTU Application software which controls the following functions:

- Software upload into the DCU for the Maintenance purposes;
- Door System Monitoring through the DCU;

- Door System Data logging retrieval from the DCU;
- Door System Troubleshooting through the DCU;
- Door System Configuration Change Parameters through the DCU:
    - Opening Time
    - Closing Time
    - Obstruction Retries (Recycles)
    - Obstruction re-open distance
    - Delay before reclosing
    - Delay before door closing
    - Door open duration for Passenger Door Enable

2 REFERENCES

This section lists the number, title, revision, date, and source of all documents referenced in this SQAP.

2.1 Standards

IEEE 730-2002	IEEE Standards for Software Quality Assurance Plans, 23 September 2002, IEEE, www.ieee.org
IEEE 1558-2004	IEEE Standard for Software Documentation for Rail Equipment and Systems
EN 50128	European Standard for Railways Application – Communications, Signaling & Processing Systems – Software for Railway Control & Protection Systems, March 2001

2.2 Project Related Documents

201111207	Software Project Management Plan
201111186	Software Configuration Management Plan, Door Control System
201111217	Software Verification and Validation Plan, Door Control System

2.3 VSRS Documents

CMM-01	Software Engineering Process (SEP), Rev.3, 2001-03-31
CMMI-02	Software Engineering Policy Manual, Rev. 1, 2008-07-12
SWI-01	Software Configuration Management (SCM) Guide, Rev.2, 2001-03-27
SWI-02	Coding Standard Guide, Rev.3, 2001-03-27
SWI-05	Inspection Guide, Rev.3, 2001-03-27
SWI-06	Code Walkthrough Guide, Rev.1, 2000-03-03
SWI-07	Lessons Learned Guide, Rev.1.0, 2000-03-03
SWI-08	SPRM Guide, Rev.2.0, 2000-03-03
SWI-09	PRM Guide, Rev.2.0, 2000-03-03
SWI-10	SRM Guide, Rev.3, 2001-03-31
SWI-16	Risk Guide, Rev.2.0, 2000-03-03
S01-01	Software Problem Report (SPR) Form, Rev.1.0, 2001-03-30
S05-01	Inspection Form, Rev.2.0, 2000-01-30
S06-01	Code Walkthrough Form, Rev.1.0, 2001-01-30
S13-01	Audit Notification, Rev.1, 1999-01-08
S13-02	Improvement Action Request (IAR) Form, Rev.1.1, 2001-01-30
S13-03	Checklist Template, Rev.1, 1999-09-29
S16-01	Risk Form, Rev.2.1, 1996-10-14
S17-01	Software Requirements Specification Template
S17-02	Software Development Plan Template
S17-04	Software Configuration Management Plan Template
S17-05	Software Configuration Items Summary Table Template
S17-06	Software Requirements Verification and Traceability Matrix Template
S17-07	Software Design Document Template
S17-08	Software Test Plan Template
S17-09	Software Test Designs Specification Template
S17-10	Software Test Procedures Specification Template
S17-12	Software User Documentation Template
S17-13	Software Verification and Validation Plan Template
S17-14	Software Verification and Validation Report Template

2.4 Other Documents

N/A

3 MANAGEMENT

This section describes the organization, tasks, and responsibilities of the project, with special emphasis on software quality assurance.

3.1 Organization

This section describes the complete VSRS internal management structure for the project. An outline of the RTD Denver EMU Door System - Project Organizational Chart (POC) is provided in Figure A-1.

- **The Engineering Manager** has the responsibility to oversee and direct the entire Engineering department, namely the Software, Electrical, Mechanical and Testing team as well the Document Control Group. The Engineering Manager has a direct communication link with the SQA to discuss any quality related concerns or issues the SQA may have.
- **The Software and Electrical Engineering Manager (SEEM)** interfaces with Engineering Manager. He is responsible for the software and electronic development. The SEEM manages the software and the electrical teams and interfaces with other managers concerning software and electrical issues.
- **The Software Department Supervisor*** ** interfaces with the Engineering Manager**.** He supervises the Software Development Team and coordinates their activities. He communicates with the other departments and supports them. The Software Department Supervisor interfaces also with the Clients and provides the technical support if needed.
- **The Mechanical Engineering Manager** interfaces with the PM and is responsible for the mechanical aspect of the project. The Mechanical Engineering Manager manages the mechanical team and interfaces with other managers concerning mechanical issues.
- **The Technical Publications and Training Manager** interfaces with the PM and is responsible for the RAMS and documentation control aspects of the project. The Engineering Services Manager manages the RAMS and the documentation control team and interfaces with other managers concerning RAMS and documentation control issues.
- **The Test and Engineering Service Manager** manages the testing team and interfaces with other managers concerning testing related issues and needs.
- **The Project Manager** coordinates activities as per the schedule and provides regular and timely communications. He also resolves conflicts across project organizations, and interfaces with project development team.
- **The Project Engineer**** has the overall responsibility of the technical aspects of the project. The Project Engineer interfaces with the customer in matters of a technical nature. In addition, the Project Engineer coordinates mechanical and electrical activities.
- **The System Engineer (SE)**** concentrates on the day-to-day management of the electrical and RAMS activities. In addition, the SE interfaces with the customer in matters of a technical nature.
- **The Software Team Leader (STL)** concentrates on the day-to-day management of the software activities; software schedules control and software technical reviews. He is also responsible for the creation of the SCMP document.
- **Software Engineer (SWE)/Designer** is responsible for software requirements & design (and associated documentation) overview and guidance; detailed design and coding; preparation of test plans, procedures, and reports; software unit testing and SCI testing prior to software release. SWE/Designer ensures that the correct configuration is in place for testing and incorporates approved changes into all the related documentation. SCM identifies all change requests and planned tests to be included in an engineering release. Software Designer then provides SCM a copy of the test reports.

- **The Software Quality Assurance (SQA)** takes part in all software development activities throughout the life cycle of the SEP process. The SQA ensures all required standards for the project will be followed. The SQA reports to the SEEM in order to discuss any quality related concerns or issues. The SQA also interfaces with the STL on a regular basis. The SQA concentrates on monitoring and controlling the daily software activities by inspecting documentation, attending project meetings and performing periodical software audits to ensure all the standards and required process elements are satisfied. The SQA works with the SEEM to follow up the findings/observations from the audits and other quality related issues for improvement. Also, SQA is a member of SCCB who will participate in the project's SCCB meetings prior to each software release.
- **The Software Configuration Manager (SCM)*** ** is responsible for the identification, control, status accounting of the software work products. The SCM releases work products to the DCG with the directives of SEEM.
- **The Document Control Group (DCG) – Technical Writer** is responsible for the receipt, distribution, and tracking of technical data associated with the project. The DCG is a service provided by the Technical Publications and Training department. The DCG releases the work products to the production department under the directives of the SCM.
- **The RAMS Engineer** concentrates on monitoring and performing periodical RAMS activities to assure progress towards specific RAMS goals and to insure that the supplied equipment will be reliable, safe, and easy to maintain.
- **The Electrical Team Leader** concentrates on the day-to-day management of the electrical activities (Hardware components design/development/technical reviews and schedule control).
- **The Mechanical Team Leader** concentrates on the day-to-day management of the mechanical activities (Mechanical components design/development/technical reviews and schedule control).
- **The Electrical Engineer*** is responsible to design and develop hardware components of the HVAC system.
- **The Mechanical Engineer/Designer** is responsible to design and develop mechanical components of the Door System.
- **The Test Engineer*** is responsible to design and develop tests for the Software/Hardware components of the RTD Denver EMU Door system.

*The Electrical Engineer assumes also the role of the Electrical Test Engineer.

**Project Engineer assumes also the role of the System Engineer.

***Software Department Supervisor acts also as the SCM.

3.2 Tasks

Table 3.1 identifies the major activities, responsibilities, and skill/ability required by the personnel and/or groups who are involved in the software project. Details of activities, responsibilities, and dependencies of the resources are presented in the Software Development Process module of the VSRS's *CMM-01: Software Engineering Process*. This plan covers the following life-cycle phases of the Software Development Process module:

- SPP-200 Software Project Planning;
- SPG-300 Software Plan Generation;
- SRD-400 Software Requirements Definition;
- SD-500 Software Development;
- SCT-600 Software Coding and Testing;
- SV-700 Software Validation;

- SDT-800 Software Development Tracking.

3.3 Roles and Responsibilities

The generic responsibilities for each task are specified in detail in the Software Development Process module of the VSRS's *CMM-01: Software Engineering Process*.

Table 3-1 identifies the specific project's organizational elements responsibilities.

Table 3-1: Resources and Responsibilities

Role	Qty	Organization	Required skill/ability	Responsibilities
VSRS Engineering Manager	1	VSRS Engineering	• Knowledge of VSRS policies processes and procedures. • Excellent planning and organizational abilities. • Well developed communication skills. • Experience in the design and implementation of electrical, mechanical software systems	• Monitors implementation of quality activities. • Oversee and direct the entire Engineering department, namely the Software, Electrical, Mechanical and Testing team as well the Document Control Group. • Direct communication link with the SQA to discuss any quality related concerns or issues the SQA may have.
VSRS Software and Electrical Engineering Manager (SEEM)	1	VSRS Engineering	• Knowledge of VSRS policies, processes and procedures. • Excellent planning and organizational abilities. • Well developed communication skills. • Experience in the design and implementation of electrical, mechanical software systems	• Interfaces with Engineering Manager • Responsible for the software and electronic development. • Manages the software and the electrical teams and interfaces with other managers concerning software and electrical issues.
VSRS Software Department Supervisor	1	VSRS Engineering	• Knowledge of VSRS policies, processes and procedures. • Excellent planning and organizational abilities. • Well developed communication skills. • Experience in the design and implementation of electrical, mechanical software systems	• Interfaces with the SEEM. • Supervises the Software Development Team and coordinates their activities. • Communicates with the other departments and supports them. • Interfaces also with the Clients and provides the technical support if needed.
VSRS Software Quality Assurance	1	VSRS Engineering	• Knowledge of VSRS policies, processes and procedures. • Experienced in software audit activities. • Experienced in inspection of software documentation. • Knowledge of FCA & PCA activities.	• Monitors implementation of quality activities. • Audits the software activities as per the project plan. • Approves project deliverables from QA perspective. • Reviews plans and deliverables for compliance with applicable standards. • Provides guidance and assistance on process matters. • Monitors implementation of SCM activities. • Audits the SCM activities as per the project plan. • Performing PCA and FCA audit.
VSRS Software Team Leader (STL)	1	VSRS Engineering	• Knowledge of VSRS processes and procedures. • Well developed communication skills. • Excellent planning and organizational abilities. • Experience in the design and implementation of electrical, mechanical and software systems	• Concentrates on the day-to-day management of the software activities. • Controls the software schedule. • Conducts the software technical reviews. • Creates and maintains the SCMP document.
VSRS Software Configuration Manager	1	VSRS Engineering	• Knowledge of VSRS policies, processes and procedures • Experienced in software audit activities. • Experienced in inspection of software documentation.	• Updated/Customizing the Software Accounting Database • Performing the Formal Baseline during the software life cycle development. • Controlling the software change on the software product.

Role	Qty	Organization	Required skill/ability	Responsibilities
				• Documenting the software change on the software product. • Producing the SVD document. • Identifying the software product to be under configuration. • Initiating and participating to the SCCB activity. • Releasing the software work product. • Performing the status accounting activities.
VSRS Project Manager	1	VSRS Project Management	• Knowledge of VSRS policies, processes and procedures. • Well developed communication skills. • Excellent planning and organizational abilities.	• Interfaces with customer and project team. • Manages commercial and financial activities. • Resolves conflicts across project organizations. • Provides regular and timely communications.
VSRS Project Engineer	1	VSRS Engineering	• Knowledge of VSRS processes and procedures. • Well developed communication skills. • Excellent planning and organizational abilities. • Experience in the design and implementation of electrical, mechanical and software systems	• Coordinates mechanical, electrical, and software activities. • Provide the Technical support to the team members. • Assures a good integration of the software and hardware. • Oversees and manages the Electrical, Mechanical and SW tasks • Ensures that the software product resulting from the development process meets the technical requirements specified in the contract. • Interfaces with the customer in matters of a technical nature.
VSRS Software Engineer	1	VSRS Engineering	• One to three years experience in the structured analysis, design and implementation of embedded software system. • Knowledge of data processing and communication technology. • Experience with C programming language and real-time Operating Systems (RTOS) • General knowledge of the MS-Windows environment. • Experience in test planning and test methodologies.	• Performs all activities on software development. • Responsible for the documentation and status of all software problem reports and maintains a System Problem Reports database. • Responsible for performing formal testing of the software product. • Develops and maintains test cases and procedures • Create the SCMP document.
VSRS Software Engineering Process Group	1	VSRS Engineering	• Organization's culture and processes knowledge. • Software Development. • Knowledge for software engineering tools.	• Provides process and tools training. • Develops and maintains software engineering processes. • Participates in software project reviews.
VSRS Software Configuration Control Board (SCCB)	5	VSRS Engineering	• VSRS SCM & SQA • PM • Project Manager • SW Designer	• Authorizes the establishment of software baselines and the identification of configuration items/units. • Represents the interests of the project manager and all groups who may be affected by changes to the software baselines. • Reviews and authorizes changes to the software baselines. • Authorizes the release of products from the software baseline library.

Note1: If staffing fluctuations occur, it will be the Engineering Manager's responsibility to oversee the replacement of personnel with appropriate skills and experience on a timely basis.

Note2: In Table 3-1 Roles and Responsibilities related to the Software are included. For more roles and responsibilities see the descriptions included in Section 3.1.

4 DOCUMENTATION

The project will follow the process and activities defined in the VSRS's CMM-01. The VSRS's CMM-01 provides the guidance governing the development, verification and validation, use and maintenance of software.

The following methods will be used to verify the adequacy of work products. These activities are part of the VSRS's CMM-01.

- All documentation, including test documentation, will be inspected as per the VSRS's SWI-05 Inspection Guide. The primary goal of the inspection process is to remove defect from work products early and efficiently in a software project; the secondary goal is to provide evidence of V&V tasks execution and task results. It is at the inspections that the documents are verified for compliance to their documentation requirements (as per section 5.1).
- Internal Reviews (see Section 6) will be performed to verify the adequacy of the documentation and its traceability.

The Table 4-1 presents the documents and the review/audit at which their adequacy will be verified.

Table 4-1: Documentation and their verification

Document Type	Quantity	Inspection	SSR/PDR	CDR	TRR	FCA	PCA	Requested by Client
SPMP	1	✓	✓				✓	✓
SQAP	1	✓	✓	✓			✓	✓
SFD	1	✓	✓	✓		✓	✓	✓
SCMP	1	✓	✓				✓	✓
SVVP	1	✓	✓	✓			✓	✓
SVVR	1	✓				✓	✓	✓
SCIST	1	✓				✓	✓	✓
SRS	per SCI	✓		✓		✓	✓	✓
SDD	per SCI	✓		✓			✓	✓
SCTP*	per SCI	✓	✓*	✓	✓		✓	✓
SCTR	per SCI	✓				✓	✓	✓
SITP	per SCI	✓	✓*	✓	✓		✓	✓
SITR	per SCI	✓				✓	✓	✓
SyTP	1	✓	✓*	✓	✓		✓	✓
SyTR	1					✓	✓	✓
SATP	1	✓	✓*	✓	✓		✓	✓
SATR	1	✓				✓	✓	✓
SRTM	1	✓				✓	✓	✓
SUD	1(for PTU only)	✓				✓	✓	✓
SVD	1	✓				✓	✓	✓

*As the Test Procedure and Test Plan are included in the same document (see Section 4.9) section related to the test planning should be provided at PDR but full document including section related to the Test Procedure will be provided at CDR.

4.1 Software Project Management Plan (SPMP)

The Software Project Management Plan (SPMP) defines the software project management responsibilities, resources, processes, and standards, to be followed for all software activities necessary to accomplish the development and maintenance of software product.

The SPMP shall be based on and comply with the requirements of IEEE Std 1058, standard for Software Project Management Plan. The SPMP shall also comply with the documentation requirements of VSRS's SPMP Template S17-02. The SPMP shall cover all software developed for this project.

4.2 System Functional Description (SFD)

The System Functional Description (SFD) describes all general requirements, operations and functions of the RTD Denver EMU Door System. It should describe the whole system, including the hardware, the housing and all the environmental requirements that have to be met.

IEEE Standard 1233-1998 "Guide for Developing System Requirements Specifications" document shall be used for SFD creation.

4.3 Software Configuration Management Plan (SCMP)

The SCMP documents the methods to be used for identifying software items, controlling and implementing changes, and recording and reporting change implementation status.

The SCMP shall be based on and comply with the requirements of IEEE Std 828, standard for Software Configuration Management Plan. The SCMP shall also comply with the documentation requirements of VSRS's SCMP Template S17-04.

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4.4 Software Verification and Validation Plan (SVVP)

The Software Verification and Validation Plan (SVVP) documents the verification and the validation tasks to determine if the developed software products are conform to their requirements, and whether the software products fulfill the intended use and user expectations.

The SVVP shall be based and comply with the requirements of IEEE Std 1012, standard for Software Verification and Validation. The SVVP shall also comply with the documentation requirements of VSRS's SVVP Template S17-13.

4.5 Software Verification and Validation Report (SVVR)

The Software Verification and Validation Report (SVVR) describes the results of the execution of the SVVP.

The SVVR shall be based and comply with the documentation requirements of VSRS's SVVR Template S17-14.

4.6 Software Configuration Items Summary Table (SCIST)

The SCIST describes each SCI.

The SCIST shall be based and comply with the documentation requirements of VSRS's SCIST Template S17-05.

4.7 Software Requirements Specification (SRS)

The Software Requirements Specification specifies all non-functional and functional requirements (for a particular software product, program or a set of programs that perform certain functions in a specific environment). It should address the basic issues of functionality, external interfaces, performance, attributes, and design constraints imposed on implementation.

The SRS shall be based and comply with the requirements of IEEE Std 830, recommended practice for Software Requirements Specification. The SRS shall also comply with the documentation requirements of VSRS's SRS Template S17-01.

4.8 Software Design Description (SDD)

The Software Design Description (SDD) depicts how the software will be structured to satisfy the functional requirements in the SRS. It describes the components and subcomponents of the software design, including databases and internal interfaces.

The SDD shall be based and comply with the requirements of IEEE Std 1016, recommended practice for Software Design Description. The SDD shall also comply with the documentation requirements of VSRS's SDD Template S17-07.

4.9 Software Test Plans

4.9.1 Software Component Test Plan (SCTP)

The Software Component Test Plan (SCTP) will be created for each of the SCIs unit and includes 3 major sections:

- **Software Component Test Planning** identifies the tests required to verify if a component satisfies its intended use.
- **Software Component Test Design Specification** described the tests design required to verify if a component satisfies its intended use.
- **Software Component Test Procedures Specification** describes the procedures that need to be executed in order to validate the design.

4.9.2 Software Integration Test Plan (SITP)

The Software Integration Test Plan (SITP) will be created for each of the SCIs and includes 3 major sections:

- **Software Integration Test Planning** identifies the tests required to verify and validate each SCI. Its content shall be based and comply with the requirements of IEEE Std 829, standard for Software Test Documentation. The ITP shall also comply with the documentation requirements of VSRS's Test Plan Template S17-08.
- **Software Integration Test Design Specification** describes the design of the tests to be performed in order to validate the requirements of each SCI.
- **Software Integration Test Procedures Specification** describes the procedures that need to be executed in order to validate the requirements of each SCI.

4.9.3 System Test Plan (SyTP)*

The System Test Plan (SyTP) will be executed at the System Level and includes 3 major sections::

- **System Test Planning** identifies the tests required to validate whether the system meets its original objectives.
- **System Test Design Specification** describes the design of the tests performed in order to validate whether the system meets its original objectives.
- **System Test Procedures Specification** describes the procedures that need to be executed in order to validate whether the system meets its original objectives.

*Please note that the SyTP will be created and executed by the System department.

4.9.4 Software Acceptance Test Plan (SATP)*

One (1) Software Acceptance Test Plan (SATP) will include 3 major sections:

- **Software Acceptance Test Planning** identifies the tests required to verify if the system satisfies its acceptance criteria
- **Software Acceptance Design Specification describes** the design of the tests performed in order to validate the acceptance criteria.
- **Software Acceptance Procedure Specification describes** the procedures of the tests performed in order to validate the acceptance criteria.

*Please note that the SATP should be created and executed by the Client.

4.10 Test Report

The Test Report provides a chronological record of relevant details about the execution of tests listed above.

The Test Reports include Software Acceptance Test Report (SATR), System Test Report (SyTR), Software Integration Test Report (SITR) as well as the Software Component Test Report (SCTR).

All Test Reports are planned to be provided individually. All references related to this documents will be listed in the Annex of the SVVR document.

4.11 Software Requirements Traceability Matrix (SRTM)

The Software Requirements Traceability Matrix (SRTM) provides a central point of reference and a starting point for an upward and downward traceability of all allocated software requirements. SFD, SRS, SDD and STP documents will be traceable.

The SRTM shall be based and comply with the documentation requirements of VSRS's SRTM Template S17-06.

There is a series of planned activities to be conducted, such as inspections, internal/external reviews (PDR, FDR/CDR, TRR, etc.). All these activities ensure adequacy of the artifact.

4.12 Software User Documentation (SUD) – For PTU only

The Software User Documentation guides the users in installing, operating, managing, and maintaining software products

The SUD should describe the required data and control inputs, input sequences, options, program limitations, and other activities or items essential for a successful execution of the software.

The SUD shall be based and comply with the requirements of IEEE Std 1063, standard. The SUD shall also comply with the documentation requirements of VSRS's Template S17-12.

4.13 Software Version Description (SVD)

The SVD establishes the baseline of the software products and describes all the changes from the previous software version(s).

The SVD shall be based and comply with the documentation requirements of the Hyundai Rotem's Software Version Description document covered by the VSRS's SVD Template.

5 STANDARDS, PRACTICES, CONVENTIONS, AND METRICS

This section identifies the standards, practices, conventions, and metrics to be applied, and states how compliance with these items is to be monitored and assured.

5.1 Documentation Standards

Table 5-1: Documentation Standards

Document Type	Applicable Standard/Template
SPMP	As per section 4.1
SQAP	IEEE 1558, IEEE 730 and VSRS S17-03: SQAP Template
SFD	As per section 4.3
SCMP	As per section 4.3
SVVP	As per section 4.4
SVVR	As per section 4.5
SCIST	As per section 4.6
SRS	As per section 4.7
SDD	As per section 4.8
SCTP	As per section 4.9.1
SITP	As per section 4.9.2
SyTP	As per section 4.9.3
SATP	As per section 4.9.4
SITR	As per section 4.10
SCTR	As per section 4.10
SyTR	As per section 4.10
SATR	As per section 4.10
SRVTM	As per section 4.11
SUD	As per section 4.12
SVD	As per section 4.13

5.2 Logic Structure Standards

The baseline for the logic structure standards is VSRS's SWI-02 Coding Standards. Compliance to VSRS's SWI-02 is monitored and assured as per the VSRS's SWI-06 Code Walkthrough Guide.

5.3 Coding Standards

The baseline for the coding standards is VSRS's SWI-02 Coding Standards. Compliance to VSRS's SWI-02 (see section 2 - C/C++ and Similar Languages) is monitored and assured as per the VSRS's SWI-06 Code Walkthrough Guide.

5.4 Commentary Standards

The baseline for the comments standards is VSRS's SWI-02 Coding Standards. Compliance to VSRS's SWI-02 is monitored and assured as per the VSRS's SWI-06 Code Walkthrough Guide.

5.5 Testing Standards and Practices

5.5.1 Testing Standards and Practices

This section identifies the tools, techniques and methodologies used for supporting testing activities. More details are provided in the SVVP Document #201111217.

Table 5-2 presents a glossary of the testing effort and its purpose. Table 5-3 presents the test activities in their typical execution order through the testing phase of the project.

Table 5-2: Glossary of Levels of Testing

Testing	Description	Purpose	Traceability
Component Testing	Testing conducted to verify the correct implementation of the design and compliance with program requirements for one software element (e.g. unit, module) or a collection of software elements (as per IEEE Std 1012).	Ensures program logic is complete and correct. Ensures all required components function as designed.	From applicable SDD design requirements, incrementally, to a test.
Integration Testing	Orderly progression of testing of incremental pieces of the software program in which software elements, hardware elements, or both are combined and tested until the entire system has been integrated to show compliance with the program design, and capabilities and requirements of the system (as per IEEE Std 1012).	Ensures that design objectives are met.	From SRS software requirements, incrementally, to a test.
System Testing	The activities of testing an integrated hardware and software system to verify and validate whether the system meets its original objectives (as per IEEE Std 1012).	Ensures that the software and hardware as complete entity complies with its operational requirements.	From each SFD incrementally, to a test.
Acceptance Testing	Testing conducted in an operational environment to determine whether a system satisfies its acceptance criteria (i.e. initial requirements and current needs of its user) and to enable the customer to determine whether to accept the system (as per IEEE Std 1012).	Ensures that customers' requirements objectives are met and that all the components are correctly included in a customer package.	From Technical Specification, incrementally, to a test.

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Table 5-3: Test Activities through the testing phase

Activities	Requirements	Design	Implementation	Test
Test Plan Generation	• System • Acceptance	• Component • Integration		
Test Design Generation		• Component • Integration • System • Acceptance		
Test Case Generation			• Component • Integration • System • Acceptance	
Test Procedure Generation			• Component • Integration • System	• Acceptance
Test Execution			• Component	• Integration • System • Acceptance

5.5.1.1 Tools

- Rational ClearQuest database for Software Problem Reporting (SPR) management,
- Rational Requisite Pro for requirements traceability tracking,
- Source Safe tool is used for version control.
- Microsoft Project for planning and project management,
- ABAK for employee time tracking,
- Microsoft Excel for metric measurements and analysis,
- Practiline Source Code Line Counter for counting source code lines.

5.5.1.2 Techniques

- Software Engineering Process (CMM-01),
- IEEE Std 1012 for Software Verification and Validation,
- IEEE Std 829 for Test Documentation.

5.5.1.3 Methodologies

- CMU/SEI-2006-008, CMMI for Development Version 1.2, CMMI Practices
- EN 50128 European Standard for Railways Application – Communications, Signalling & Processing
- Systems – Software for Railway Control & Protection Systems, March 2001 – for SIL 2

5.6 SQA Metrics

Metric actions are performed to gather historical data to manage and estimate schedules, tasks, resources, tools and costs during the Project and for further Projects. Metric collection will be analyzed and compiled to get statistical data. The following section summarizes the metrics collected, the reason why they are collected, the method of collection, the frequency of collection, the data representation and the responsibilities for collection, reporting and analysis.

5.6.1 Effort

Table 5-4: Metrics collection

Action	Description	Reason	Method	Representation	Responsible	Frequency
Effort	Number of hours spent for each item of the documentation, engineering and management.	Assess if calculated effort corresponds to the real effort. Provide historical data.	Each individual fills in his time sheet and this will be entered in the Projects database. Each item is entered using the codes on the Project-planning sheet.	Excel file	**Collection:** Each team member. **Reporting:** Project Engineer and Project Manager. **Analysis:** Project Engineer and Project Manager.	At least weekly
Project Open Items List	List of all open items concerning the general project level to resolve.	Prevent any action to be forgotten, checking and updating resources.	During the project meetings all open items are discussed and new items are added to the list.	Excel file	**Collection , Reporting and Analysis:** Project Engineer and Project Manager	Weekly
SPRs	List of all open items concerning software issues to be resolved.	Prevent any software action to be forgotten.	The list is kept up continuously during development and testing.	Excel file	**Collection**: Every Software Engineer. **Reporting and Analysis:** Project Engineer and Project Manager	Upon modifications
Requirem ents	Collecting and distributing all required Project requirements.	Assuring that the product requirements are up to date.	Emails, meetings and conference calls with the client.	Implementation in the SFD, SRS and the quality documents.	**Collection, Reporting and Analysis:** Project Engineer and Project Manager.	Upon modifications

6 REVIEWS AND AUDITS

This section explains and details the reviews and audits activities.

Every audit shall be performed at a minimum as defined in the VSRS's CMM-01, and if required, the Auditor can perform other audits as per:

- The result of previous audits.
- Relative importance of the software activity.
- The extent of revisions or activities performed between the plan audits.
- The extent of change in the software activity.
- The number of activities between two planned audits.

Every audit/review inspects the related document as defined in the VSRS's CMM-01. A report is then produced at the end of activity that includes a list of non-conformities and notes. An Improvement Action Request is produced for every documents or activity that contains non-conformity.

6.1 Technical Reviews

VSRS shall conduct any technical reviews with RTD Denver as required, such as Preliminary Design Review, Final Design Review, etc.

Prior to each review with RTD Denver, VSRS shall perform a series of internal reviews as per CMM-01 to ensure that all the requirements/mandates are satisfied. These reviews specified in the following subsections shall cover all technical aspects, e.g., conceptual design, preliminary design, detailed design and final design and implementation.

6.1.1 System Functional Description Review (SFDR)

The System Functional Description Review (SFDR) is held to ensure the adequacy of the system description and requirements stated in the SFD. Especially the correctness of all interfaces will be checked. The SFDR is conducted as per the SQA Guide document SWI-13

6.1.2 Software Requirements/Specification Review (SSR)

The SSR is held to ensure the adequacy of the requirements stated in the SRS. The SSR is conducted as per the SQA Guide document SWI-13 and CMM-01 module SRD-400. The details of the actions to be performed are documented in step SRD-460.

6.1.3 Architecture design review (PDR)

The PDR is held to evaluate the technical adequacy of the preliminary design (also known as the top-level – Architecture Design Review) of the software as depicted in the preliminary SDD. The PDR is conducted as per the SQA Guide document #SWI-13 and the CMM-01 module SD-500. The details of the actions to be performed are documented in step SD-550.

6.1.4 Detailed design review (CDR)

The CDR (also known as detailed design review) is held to determine the acceptability of the detailed software designs as depicted in the SDD in satisfying the requirements of the SRS. The CDR is conducted as per the SQA Guide document #SWI-13 and scheduled in the CMM-01 module SD-500. The details of the actions to be performed are documented in step SD-570.

6.1.5 Software Verification and Validation Plan Review (SVVPR)

The SVVPR is held to evaluate the adequacy and completeness of the verification and validation methods defined in the SVVP. The VVPR is conducted as per the SQA Guide document #SWI-13 and scheduled in the CMM-01 module SDT-800. The details of the actions to be performed are documented in step SDT-840.

6.2 Product Audits

6.2.1 Functional audit (FCA)

The Functional audit is held prior to the software delivery to verify that all requirements specified in the SRS have been met. The FCA is conducted as per the SQA Guide document #SWI-13 as well as the SCM Guide document #SWI-01 and scheduled in CMM-01 module SV-700. The FCA is further detailed in step SV-730 as well as in the project's SCMP document #201111186

6.2.2 Physical Audit

The Physical Configuration Audit is held to verify that the software and its documentation are internally consistent and are ready for delivery.
The PCA is conducted as per the SQA Guide document #SWI-13 as well as the SCM Guide document #SWI-01 and scheduled in CMM-01 module SV-700. The PCA is further detailed in step SV-730 as well as in the project's SCMP document #201111186

6.3 In-Process Audits

In-Process audits are conducted as per the CMM-01 to assess the compliance of the software development and maintenance process activities against the standards and practices defined for the project. Any nonconformance found during the audits will be recorded and reported to the STL or to the SEEM. It is the STL's responsibility together with the SEEM to coordinate the effort among the involved parties and to decide the proper timing to take corrective actions on the findings.

Note that VSRS's CMM-01 guide requires that the audits are held to ensure adherence with the project plans, such as SPMP, SQAP, SCMP and SVVP, in different phases of software development life cycle.

The audit on the Software Project Planning (SPP-200) activities is conducted as per the SQA Guide document #SWI-13 and scheduled in CMM-01 module SPP-200. The details of the actions to be performed are documented in step SPP-260.

The audit on the Software Plan Generation (SPG-300) and Software Requirements Definition (SRD-400) activities is conducted as per the SQA Guide document #SWI-13 and scheduled in CMM-01 module SRD-400. The details of the actions to be performed are documented in step SRD-470.

The audit on Software Development (SD-500) activities is conducted as per the SQA Guide document #SWI-13 and scheduled in CMM-01 module SD-500. The details of the actions to be performed are documented in step SD-590.

The audit on the Software Coding and Testing (SCT-600), Software Validation (SV-700) and Configuration Management activities is conducted as per the SQA Guide document #SWI-13 and scheduled in CMM-01 module SV-700. The details of the actions to be performed are documented in step SV-760.

6.4 Managerial Reviews

An independent quality assurance group (independent to the SQA) performs one audit on SQA activities during the project. The audit is conducted as per the SQA Guide document #SWI-13 and scheduled in the CMM-01 module SDT-800. The details of the actions to be performed are documented in step SDT-840.

Any nonconformance/findings found during the audits will be recorded and reported to the STL or SEEM. It is the STL's responsibility together with the SEEM to coordinate the effort among the involved parties and to decide the proper timing to take correction actions on the findings.

The findings will be analyzed and evaluated to find if there will be any impact on the project and the development process. If so, the SQAP shall be updated to address the project-related findings and the implementation of their correction actions. For the process-related findings, VSRS will address them as continuous process improvement.

6.4.1 Software Configuration Management Plan Review (SCMPR)

The Software Configuration Management Plan Review (SCMPR) is held to evaluate the adequacy and completeness of the configuration management methods defined in the SCMP. The SCMPR is conducted as per the SQA Guide document #SWI-13 and the CMM-01 module SDT-800. The details of the actions to be performed are documented in step SDT-840.

6.4.2 Post-implementation plan review

This review is held at the conclusion of the project to assess the development activities and to provide recommendation for process improvements. The lessons learned session is conducted as per the Lessons Learned Guide document #SWI-07 and scheduled in CMM-01 module SV-700. The details of the actions to be performed are documented in step SV-780.

Any lesson learned information will be recorded and reported to the SEEM. It is the SEEM's responsibility to coordinate the improvement effort and to decide the proper timing to apply the lesson learned to the future projects.

7 TEST

This section is not applicable to this plan. Tests will be described in the SVVP document.

8 PROBLEM REPORTING AND CORRECTIVE ACTION

VSRS is determined to deliver quality software products for all its customers. As such, problem reporting and implementation of the solutions to those problems will be communicated to VSRS senior management. It is the senior manager's responsibility to sponsor these kinds of activities.

8.1 Improvement Action Request (IAR)

The VSRS's *S13-02: Improvement Action Request (IAR) Form* and its database shall be used to document and track the Improvement Action Request on a software activity, documentation or work product during the project. The IAR Form indicates the non-conformance, standard requirements that shall be followed, the immediate improvement required, the root cause of the non-conformance and the permanent improvement to be implemented to eliminate the recurrence of that type of non-conformance.

The IAR database shall also be used to track and monitor progress and follow up of the corrective action issues. The IAR database is also used to perform metric and statistical data to analyze the trends of the non-conformances and for future projects.

Improvement Action Request is performed as per the *SWI-13: SQA Guide*. Responsibilities are as per the SWI-13.

8.2 Software Problem Report

The VSRS's *S01-01: Software Problem Report (SPR) Form* and database shall be used to control, track and monitor issues found after a software work product has been baselined or released. Software problem reporting and tracking is performed as per the VSRS's *SWI-01: SCM Guide*. Responsibilities are as per the VSRS's SWI-01.

8.3 Document Inspection

The VSRS's *S05-01: Inspection Form* and database shall be used to document and track to closure issues found on the software documentation. The resulting inspection forms provide evidence of V&V activities execution. Document inspections are performed as per the VSRS's *SWI-05: Inspection Guide* where the VSRS's SQA acts as the inspection leader.

8.4 Code Walkthrough

The *S06-01: Code Walkthrough Form* and database shall be used to document and track to closure the software code issues. The resulting inspection form provides evidence of V&V execution. Code walkthroughs are performed as per the *SWI-06: Code Walkthrough Guide* where the SQA acts as the inspection leader.

9 TOOLS, TECHNIQUES, AND METHODOLOGIES

This section presents the tools, techniques and methodologies used for supporting activities described in this plan.

9.1 Tools

- Microsoft Project shall be used to support project scheduling and monitoring effort metrics on the project.
- QS database shall be used to track process, product non-conformance and perform statistical data analysis.
- ClearQuest - SPR database shall be used to control, track and monitor issues found after a software work product has been baselined or released.
- Rational RequisitePro is used for Requirement Traceability.
- Inspection & Code Walkthrough database shall be used to monitor defects in documentation and software code.
- E-mail system shall be used for internal communication.
- Quality Record and Historical Folder shall be used to archive documentation and metrics data.
- S05-02 to S05-15 Documentation Inspection Checklist Template
- S13-01 Audit Notification Form
- S13-02 IAR Form
- S13-03 to S13-21 Audit Checklist Template
- S13-22 Audit Report Template

9.2 Techniques

9.2.1 CMM-01 (Software Engineering Process)

Software Engineering Process (SEP) is a framework for development of software projects at VSRS. The SEP follows rules, requirements, and activities required in the CMM for Software, level 2, from the Software Engineering Institute (SEI).

The SEP establishes a common process framework that defines, implements, and supports the VSRS organization by specifying the methods, people and tools necessary to achieve projects and software products with the lowest cost, best delivery time and the lowest rates of defects.

The SEP delineates the framework that shall be applied in any research, design, development, maintenance or acquisition of software products by VSRS. The SEP provides a tool for VSRS management to ensure that the people, products and processes involved, are supported by the organization and a set of methods and tools capable of meeting the standards of performance required by company objectives.

Personnel must apply a predictable engineering process to design, develop, subcontract or maintain software products.

9.2.2 CMMI-02 (Software Engineering Policy Manual)

This document establishes management expectations for developing and sustaining a software engineering process capability within VSRS.

9.2.3 SWI-01 (Software Configuration Management Guide)

This document describes the Software Configuration Management process used by VSRS to define and identify software baselines along with their components and documentation, to control changes to those baselines, to verify the conformance of those baselines to the original documentation and/or to the documented changes made to them and to maintain records of the baselines status. This guide

defines outlines to all users and ensures that different individuals can repeat a set of auditable activities with appropriate diligence.

9.2.4 SWI-02 (Software Coding Standard)

This document describes the coding standard for software programs. The purpose of this coding standard is to improve maintainability, re-usability, and source-code quality.

9.2.5 SWI-05 (Inspection Guide)

This document describes the methods, resources, and the process to be used by VSRS to perform an Inspection to remove defects on a software document early and efficiently during a software project. This guide defines outlines to all users and ensures that a set of auditable activities can be repeated by different individuals with appropriate diligence.

9.2.6 SWI-07 (Lessons Learned Guide)

This document describes the methods, resources, and the process to be used by VSRS to perform the Lessons Learned activities at the end of a software project. This guide defines outlines to all users and ensures that a set of auditable activities can be repeated by different individuals with appropriate diligence.

9.2.7 SWI-08 (SPRM Guide)

This document describes the methods, resources, and processes to be used by VSRS to perform a Software Project Review Meeting (SPRM) on a software project. This guide defines outlines to all users and ensures that a set of auditable activities can be repeated by different individuals with appropriate diligence.

9.2.8 SWI-09 (PRM Guide)

This document describes the methods, resources, and processes to be used by VSRS to perform a Project Review Meeting (PRM) on a project. This guide defines outlines to all users and ensures that a set of auditable activities can be repeated by different individuals with appropriate diligence.

9.2.9 SWI-10 (SRM Guide)

This document describes the methods, resources, and processes to be used by VSRS to perform a Senior Review Meeting (SRM) on a software project. This guide defines outlines to all users and ensures that different individuals can repeat a set of auditable activities with appropriate diligence.

9.2.10 SWI-13 (SQA Guide)

This document describes the methods, resources, and processes to be used by VSRS to conduct and perform Software Quality Audit, Software Review and Configuration Audit to evaluate the status, the adequacy, and the effectiveness of the Quality Software activities conducted by VSRS. This guide defines outlines to all users and ensures that a set of auditable activities can be repeated by different individuals with appropriate diligence.

9.2.11 SWI-16 (Risk Guide)

This document describes the methods, resources, and the process to be used by VSRS to evaluate and identify the risks as early as possible and includes concepts on how to adjust the development strategy to mitigate those risks that can affect a software project. This guide defines outlines to all users and ensures that a set of auditable activities can be repeated by different individuals with appropriate diligence.

9.2.12 EWI-003 (Project Folder, Naming and Permission):

This document describes a project folder structure and naming methodology for the Engineering folder.

9.2.13 QSP-14 (Handling, Storage, Packaging, Preservation and Delivery)

This document describes the methods, resources, and the process to be used by VSRS to assure that products delivered to the customer are handled, stored, packaged, and preserved.

9.2.14 SXX-XX

VSRS's Templates and Forms.

9.3 Methodologies

IEEE 730-2002	IEEE Standards for Software Quality Assurance Plans, 23 September 2002, IEEE, www.ieee.org
IEEE 1558-2004	IEEE Standard for Software Documentation for Rail Equipment and Systems
CMU/SEI-2006-008	CMMI for Development Version 1.2, CMMI Practices
EN 50128	European Standard for Railways Application – Communications, Signaling & Processing Systems – Software for Railway Control & Protection Systems, March 2001 – for SIL 2

10 MEDIA CONTROL

The methods and facilities to be used to identify the media for each intermediate and deliverable computer work product and the documentation required to store the media, including the copy and restore process are documented in the SCMP document #201111186

The methods and facilities to be used to protect computer program physical media from unauthorized access or inadvertent damage or degradation during all phases of the software life cycle are governed by QSP-S14-02 and managed by VSRS's Management Information Service group (outside the scope of this SQAP).

11 SUPPLIER CONTROL

This section is not applicable to this plan. There is no software product acquired from a supplier.

12 RECORDS COLLECTION, MAINTENANCE, AND RETENTION

12.1 Records Collection

Data management will be performed as per VSRS's *SWI-01: SCM Guide*.

During the life cycle of the project all work products will be kept in the Project Folder, identified and accessible to personnel involved in the project. The folder structure, procedures and access rights are as per VSRS's *EWI-003: Project Folder, structure, naming and permissions*.

The SQA documentation, inspections, reviews and audits will be retained in the Software Quality Database and Software Quality Record.

12.2 Records Maintenance

The SCM documentation, baseline and release will be retained in the SCM library as documented in the SCMP document #201111186 and in the VSRS's *ReleasedToPublic* folder.

12.3 Records Retention

The Record Retention period will be up to 10 years after the guaranties period.

13 TRAINING

The team members are mostly trained but any missing skill will be addressed with adequate training during the project life cycle.

14 RISK MANAGEMENT

Risks management is performed as per the VSRS's *SWI-16: Risk Guide*, using the VSRS's *S16-01 Risk Form*. Risks are assessed periodically at the Software Project Review Meetings (SPRM) as per the C*MM-01: Software Engineering Process* and the VSRS's *SWI-08: SPRM Guide*, and the VSRS's *SWI-09: PRM Guide*, and VSRS's *SWI-10: SRM Guide*.

15 SQAP MAINTENANCE

The Revision List must be fulfilled with the current revision number, the released date, the author and a detailed description of all changes from the previous release of the SQAP. The related Engineering Change Document (ECD) number must be identified when changes apply to an approved baselined SQAP. More details about the ECD process are provided in the SCMP document #201111186.

16 GLOSSARY

Definitions and acronyms listed below establish meaning in the context of this SQAP.

16.1 Definitions

Commercial-Off-The-Shelf (COTS): Commercially available software that is readily available through retail and wholesale sources.

Configuration Item (CI): An aggregation of hardware, software, or both, that is designated for configuration management and treated as a single entity in the configuration management.

Configuration Management (CM): A discipline applying technical and administrative direction and surveillance to: identify and document the functional and physical characteristics of a configuration item, control changes to those characteristics, record and report change processing and implementation status, and verify compliance with specified requirements.

Controlled Document: A baseline document upon which subsequent development is based.

Documentation: (1) A collection of documents on a given subject. (2) Any written or pictorial information describing, defining, specifying, reporting, or certifying activities, requirements, procedures, or results. (3) The process of generating or revising a document. (4) The management of documents, including identification, acquisition, processing, storage, and dissemination.

External Interface: All interfaces (shared boundaries) between a software product and its environment. The human and application interfaces define how the functions of the software product are invoked, whereas the communications, hardware, and other interfaces specify the software product's usage of the system components (products).

Inspection: A static analysis technique that relies on visual examination of development products to detect errors, violations of development standards, and other problems. Types include code inspection and design inspection.

Milestone: A scheduled event for which a project member or manager is held accountable and that is used to measure progress; for example, a formal review, issuance of a specification, or product delivery.

Process: (1) A sequence of steps performed for a given purpose; for example, the software development process. (2) An executable unit managed by an operating system scheduler. (3) To perform operations on data.

Quality Assurance: A planned and systematic set of all actions necessary to provide adequate confidence that the item or product conforms to the established technical and quality requirements.

Requirement: A condition or capability that must be met or possessed by a system or system component to satisfy a contract, standard, specification, or other formally imposed document. The set of all requirements forms the basis for subsequent development of the system or system component.

Review: A process or meeting during which a work product, or set of work products, is presented to project personnel, managers, users, customers, or other interested parties for comment or approval. Types include code review, design review, formal qualification review, requirements review, test readiness review.

Software: Computer programs, procedures, and possibly associated documentation and data pertaining to the operation of a computer system.

Software Configuration Item (SCI): A single software component or an aggregation of software, typically an aggregate of two or more software components that is designated for configuration management and treated as a single entity in the configuration management.

Software Designers: Trained professionals involved in the engineering of software products (Analysis/Design/Implementation), and familiar with VSRS's Quality System Procedures.

Software Engineering: The systematic approach to the development, operation, maintenance, and retirement of the software.

Software Item: A software product, or a logically separable part of it.

Software Product: (1) The complete set of computer programs, procedures, and possibly associated documentation and data designated for delivery to a user. (2) Any of the individual items in (1).

Software Test Report (STR): A document summarizing testing activities and results. It also contains an evaluation of the corresponding test items.

System: A collection of components organized to accomplish a specific function or set of functions.

System Software: Software designed for a specific computer system or family of computer systems to facilitate the operation and maintenance of the computer system and associated programs, for example, operating systems, compilers and utilities. (contrast with application software.)

Test: A set of one or more test cases and test procedures.

Test Case: A set of test inputs, execution conditions, and expected results developed for a particular objective, such as to exercise a particular program path or to verify compliance with a specific requirement.

Test Procedure: Detailed instructions for the set up, operation, and evaluation of results of a given test.

User: Person or application interacting with the software product.

User Documentation: Documentation conveying to the end user of product instructions for using the product to obtain desired results. Software User Documentation (SUD) is a part of the user documentation (contrast with product documentation).

Validation: The process of evaluating a system or component during or at the end of the development process to determine whether it satisfies specified requirements.

Verification: (1) The process of evaluating a system or component to determine whether the products of a given development phase satisfy the conditions imposed at the start of that phase. (2) Formal proof of program correctness.

Version: Labeling of a configuration item, either electronically or manually, whenever this item changes.

16.2 Acronyms

CDR	Critical Design Review
CDRL	Contract Data Requirements List
CI	Configuration Item
CM	Configuration Management (Manager)
COTS	Commercial-Off-The-Shelf
DCG	Document Control Group
FCA	Functional Configuration Audit
IAR	Improvement Action Request
IEEE	Institute of Electrical and Electronics Engineers
FAI	First Article Inspection
PCA	Physical Configuration Audit
PDR	Preliminary Design Review
PM	Program Management (Manager)
POC	Project Organizational Chart
PRM	Progress Review Meeting
PTE	Portable Test Equipment
QS	Quality System
SATP	Software Acceptance Test Plan
SATR	Software Acceptance Test Report
SCCB	Software Configuration Control Board
SCI	Software Configuration Item
SCM	Software Configuration Management (Manager)
SCMP	Software Configuration Management Plan
SCMPR	Software Configuration Management Plan Review
SCTP	Software Component Test Plan
SCTR	Software Component Test Report
SDD	Software Design Description
SEEM	Software and Electrical Engineering Manager
SEP	Software Engineering Process
SEPG	Software Engineering Process Group
SFD	System Functional Description
SFDR	System Function Description Review
SIL	Safety Integrity Level
SITP	Software Integration Test Plan
SITR	Software Integration Test Report
SPR	Software Problem Report
SPMP	Software Project Management Plan
SPRM	Software Project Review Meeting
SQA	Software Quality Assurance
SQAP	Software Quality Assurance Plan
SQAPR	Software Quality Assurance Plan Review
SRM	Senior Review Meeting
SRS	Software Requirements Specification
SSR	Software Specifications Review
STDS	Software Test Designs Specification
STL	Software Team Leader
STP	Software Test Plan
STPS	Software Test Procedures Specification
STR	Software Test Report
SUD	Software User Documentation
SVV	Software Verification and Validation
SVVP	Software Verification and Validation Plan
SVVR	Software Verification and Validation Report
SVVPR	Software Verification and Validation Plan Review

SyTP	System Test Plan
SyTR	System Test Report
SVD	Software Version Description
SW	Software
V&V	Verification and Validation
VSRS	Vapor Stone Rail Systems Inc.
WBS	Work Breakdown Structure

APPENDIX A: PROJECT ORGANIZATIONAL CHART

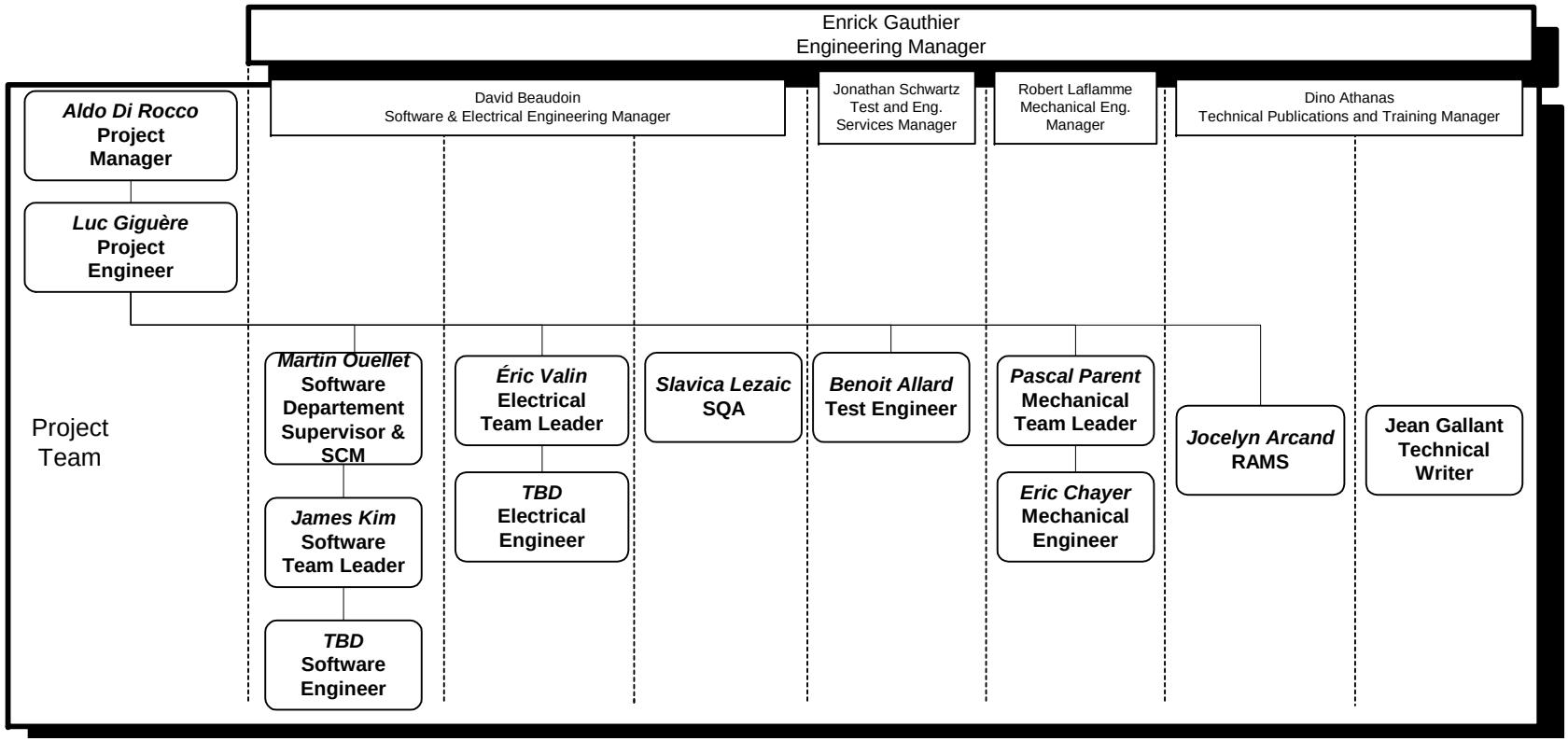

Figure A-1: Project Organizational Chart